

Automated segmentation of epidermis in high-frequency ultrasound of pathological skin using a cascade of DeepLab v3+ networks and fuzzy connectedness

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ABSTRACT

This study proposes a novel, fully automated framework for epidermal layer segmentation in different skin diseases based on 75 MHz high-frequency ultrasound (HFUS) image data. A robust epidermis segmentation is a vital first step to detect changes in thickness, shape, and intensity and therefore support diagnosis and treatment monitoring in inflammatory and neoplastic skin lesions. Our framework links deep learning and fuzzy connectedness for image analysis. It consists of a cascade of two DeepLab v3+ models with a ResNet-50 backbone and a fuzzy connectedness analysis module for fine segmentation. Both deep models are pre-trained on the ImageNet dataset and subjected to transfer learning using our HFUS database of 580 images with atopic dermatitis, psoriasis and non-melanocytic skin tumors. The first deep model is used to detect the appropriate region of interest, while the second stands for the main segmentation procedure. We use the softmax layer of the latter twofold to prepare the input data for fuzzy connectedness analysis: as a reservoir of seed points and a direct contribution to the input image. In the experiments, we analyze different configurations of the framework, including region of interest detection, deep model backbones and training loss functions, or fuzzy connectedness analysis with parameter settings. We also use the Dice index and epidermis thickness to compare our results to state-of-the-art approaches. The Dice index of 0.919 yielded by our model over the entire dataset (and exceeding 0.93 in inflammatory diseases) proves its superiority over the other methods.

1. Introduction

The fast development of ultrasound (US) imaging techniques in high frequency (HFUS) has opened up new opportunities for its medical application. Being non-invasive, convenient, and safe, it is gaining popularity and interest in dermatology, oncological-dermatology, ophthalmology, cosmetology, and esthetic medicine, while its clinical usage is still expanding (Shung, 2009). The ultrasound at frequencies of 28, 50, or 75 MHz is now commonly used in medical practice (Bhatta et al., 2018), and the > 100 MHz probes are constantly designed (Fei et al., 2016). At the expense of a shallower penetration depth, the higher frequency of the US probe improves the spatial resolution of the acquired images (Shung, 2009). Thanks to it, the superficial structures like the fat layer, the muscle layer, blood vessels, hair follicles, and skin appendages are clearly visible (de Oliveira Barcaui et al., 2016; Bezugly

et al., 2020; Czajkowska et al., 2021). The availability of HFUS for medical applications reduces the number of biopsies since it may be helpful in the determination of skin tumor depth and tumor prognosis (Shung, 2009; Heibel et al., 2020; Bezugly et al., 2020). It helps in therapeutic decisions and surgical planning. Kleinerman et al. (2012) demonstrates its potential in differentiation between melanoma, benign nevi, and seborrheic keratoses, along with monitoring of inflammatory conditions and photodamage. As reported by Heibel et al. (2020), HFUS is a reliable method with perfect intrahand interreproducibility for the measurement of melanoma depth in vivo and may enable single-step surgical excision. For basal-cell carcinoma (BCC), the spatial extent of the lesion provided by, i.e., US data analysis, is a crucial parameter for the surgical excision (Sciolla et al., 2017).

The availability of HFUS has opened up new opportunities in inflammatory skin disease diagnosis and treatment monitoring by

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analyzing the skin layers reliably, accurately, and with reduced time. The treatment effect in atopic patients observed using HFUS (20 MHz) was, as first, described by Polńska et al. (2015), whereas Lin et al. (2015) described the potential of an 18 MHz probe for the early diagnosis of psoriasis arthritis. The presence of a subepidermal low echogenic band (SLEB) below the echo entry (epidermis layer) in HFUS may indicate the presence of inflammatory dermatoses. Its thickness correlates to the severity of skin lesion (Polńska et al., 2015; Czajkowska et al., 2021). The SLEB presence in the visually healthy skin enables skin lesion differentiation. The skin thickness and stiffness measurements are recognized as a valuable supplement to clinical skin assessment in systemic sclerosis (Chen et al., 2020). Epidermal atrophy (thinning) as a result of topical corticosteroid treatment in patients with atopic dermatitis (AD) can be monitored and well documented using both optical coherent tomography (OCT) and HFUS (Byers et al., 2018). In all the mentioned applications, a robust epidermis segmentation is an essential first step to further analysis. The detection of changes in epidermis thickness, shape, and enhancement measured in a reliable and repeatable way may be a vital contribution to inflammatory and neoplastic skin lesions diagnosis and treatment monitoring.

The development of imaging techniques is usually followed by the fast progress of related image processing methods. First, Pereyra et al. (2012) described the segmentation of skin lesions in a 25 MHz US using a generalized Rayleigh mixture model. Next, Gao et al. (2013) presented an automatic skin segmentation algorithm in US images to evaluate the skin toxicity in radiotherapy. The active contour-based algorithm described there was outperformed by Sciolla et al. (2015, 2018) and Czajkowska et al. (2020) focusing on the local image analysis. Sciolla et al. (2015) introduced the level set method to the epidermis and dermis segmentation. Their first work in HFUS analysis was followed by the study on the segmentation of both skin layers (Sciolla et al., 2018) and skin tumors (Sciolla et al., 2017) using a hybrid geodesic active contour. A new, alternative model, based on diffuse interface phase field for skin tumor segmentation, was introduced by Nguyen et al. (2019) outperforming the level set idea. The fuzzy clustering-based active contour model for skin layers (epidermis and SLEB) was described by Czajkowska et al. (2020), whereas Marosán et al. (2021) introduced the skin layer segmentation based on the active contour model for automated detection of seed points for skin tumor regions. Three-dimensional graph-based skin layer segmentation in OCT was developed by Srivastava et al. (2018).

All the methods above are described as fast, accurate, and robust. However, their application is limited to small datasets, and their ability to analyze large databases with various skin lesions has not been verified. On the other hand, deep learning-based methods are now widely used in image segmentation, mainly through convolutional neural networks (CNNs), consisting of different layers responsible for dedicated data processing (Hesamian et al., 2019). The employment of graphical processing units (GPUs) reduces both the training and execution time to reasonable limits (Haque and Neubert, 2020). The introduction of the fully convolutional network (FCN) by Long et al. (2015) enabled the deep-learning dense pixel-wise semantic image segmentation.

The U-Net architecture, likely the most widely used model in medical image segmentation (Ronneberger et al., 2015), was applied to skin layer segmentation in HFUS by Czajkowska et al. (2021). The SegUNet architecture involving fuzzy C-means (FCM) clustering for preprocessing yielded a Dice index of 0.874 for epidermis segmentation. The SegUNet outperformed the other models over a published database¹ including 380 images of patients with AD or psoriasis: the CE-Net, developed by Gu et al. (2019) for 2D OCT segmentation, the original U-Net implementation (Ronneberger et al., 2015), and the SegNet architecture (Badrinarayanan et al., 2017; Khagi and Kwon, 2018). The U-Net-based

approach was also efficiently implemented by Oskal et al. (2019) for epidermal tissue segmentation in the whole slide histopathological image analysis.

Due to the similar appearance of skin or retina layers in OCT, adequate deep learning-based concepts were also applied for segmentation in this modality. Kepp et al. (2019) introduced densely-connected (DC) blocks to the original U-Net architecture. Gu et al. (2019) proposed the CE-Net approach, being inspired by the Inception ResNet (Szegedy et al., 2017). The CE-Net extends U-Net with dense atrous convolution (DAC) blocks extracting enriched feature representations through multiscale atrous convolutions. It outperformed the approach provided by Roy et al. (2017), which in turn employed the FCN. The U-Net architecture, followed by the Savitzky-Golay filter and Fourier domain filtering, was successfully applied by del Amor et al. (2020) to the epidermis and follicular structures. It yielded the average Dice index of 0.83 over 270 human skin OCT images.

Due to the problem with the access to the training data, the models above mostly lack the ability to generalize knowledge to the unknown clinical data (acquired in different medical fields, conditions, acquisition protocols, etc.) Therefore, their application to clinical practice is difficult and is likely to fail. Hence, different ideas were introduced, e.g., semi-supervised learning, transfer learning, learning from noisy labels, and learning from computer-generated labels (Karimi et al., 2020). Transfer learning (TL) is very common in medical practice. van Opbroek et al. (2015) reported that TL improves supervised image segmentation across imaging protocols in medical applications. It employs a model pre-trained in a source domain and then fine-tunes it to the target task (Karimi et al., 2020). The transferred data are mostly the network parameters. One possible strategy for re-training is to freeze the filters of the encoder (feature extraction) section and train only the decoder part. Morid et al. (2021) provided the summary of methods employing TL to medical image analysis using the CNNs trained on the non-medical ImageNet dataset (ImageNet, 2021). According to this review, the Inception models were most frequently used in ultrasound. A modified Inception model – the Xception uses depth-wise separable convolutions to involve spatial dimension and channel dimension of the image separately during training, with slightly better performance on ImageNet (Chollet, 2017). Other common models re-trained for medical tasks are the AlexNet (Krizhevsky et al., 2012), VGG-16 (Simonyan and Zisserman, 2014), and ResNet-50 (He et al., 2016). The latter utilizes intermediate normalization layers to avoid vanishing and exploding gradients (Kensert et al., 2019). Finally, the encoder-decoder DeepLab v3+ architecture (Chen et al., 2018), refining the segmentation results along target boundary, was described by Liu et al. (2021) as applicable to medical image segmentation.

Image segmentation frameworks quite often combine neural networks with fuzzy logic. Özyurt et al. (2020) introduced FCM-based segmentation for textural feature extraction followed by the CNN in the lung CT analysis. The FCM-based preprocessing was also described by Czajkowska et al. (2021) for skin layer segmentation in HFUS. Hsu et al. (2020) proposed a convolutional fuzzy neural network for image classification. A convolutional fuzzy min-max neural network introduced to handle the online learning problem for image classification was shown by Chavan and Nandedkar (2020). They used convolutional and pooling layers to extract the feature vector from the image and introduced fuzzy sets inspired by the fuzzy min-max neural network to the classification layer. Recently, Bai et al. (2020) proposed a fuzzy connectedness-based improvement to pulse-coupled neural networks. As the authors claim, integrating prior knowledge of fuzzy objects into the neural network enables robust and accurate image segmentation. A fuzzy information-guided fully convolutional network for breast US image segmentation was presented by Huang et al. (2019) with a mean intersection-over-union (IoU) of 0.8047.

The latest research in HFUS analysis reported that an accurate and reliable segmentation of skin layers might be useful in diagnosing and monitoring both inflammatory and neoplastic skin diseases (Sciolla

¹ <https://data.mendeley.com/datasets/5p7fjxt7vs/draft?a=fe8f558b-4aa1-4753-a366-8a5e3bfa7ec8>

et al., 2018; Byers et al., 2018; Heibel et al., 2020; Czajkowska et al., 2021; Marosán et al., 2021). The common aspect of their needs is the epidermal layer analysis. It is vital for the monitoring of epidermal atrophy, inflammatory conditions, photodamage, and surgical planning. To face this need, in this study, we aim to develop an automated method for epidermal layer segmentation in HFUS images (75 MHz). Since different methods addressed single medical problems (Sciolla et al., 2017, 2018; Marosán et al., 2021; Czajkowska et al., 2021), they can fail in different diseases, even applied to the same modality. Therefore, our goal is to develop the automated method for extracting the epidermal layer in different inflammatory and neoplastic skin diseases, as shown in Fig. 1. We find this approach the first application of deep learning to this broader spectrum of skin diagnosis. The recent works were limited to only neoplastic (Sciolla et al., 2017) or inflammatory (Czajkowska et al.,

2020, 2021) diseases.

Our fully-automated segmentation framework incorporates a cascade of two deep convolutional neural networks pre-trained on ImageNet, followed by fuzzy connectedness (FC) analysis for fine segmentation. Two instances of the DeepLab v3+ network enable robust region of interest (ROI) extraction, FC seed point determination, and feature extraction, whereas the FC mostly reduces false-negative areas. Our approach is the first application of a pre-trained DeepLab v3+ to HFUS images. The fuzzy connectedness-based adjustment with a dedicated procedure for the seed points (seed region) selection is another contribution of the study, yielding a noticeable increase in segmentation accuracy. The method was verified with a database consisting of 580 HFUS images of AD or psoriasis (380) and neoplastic diseases (200), all delineated by an expert. The obtained results demonstrate that the

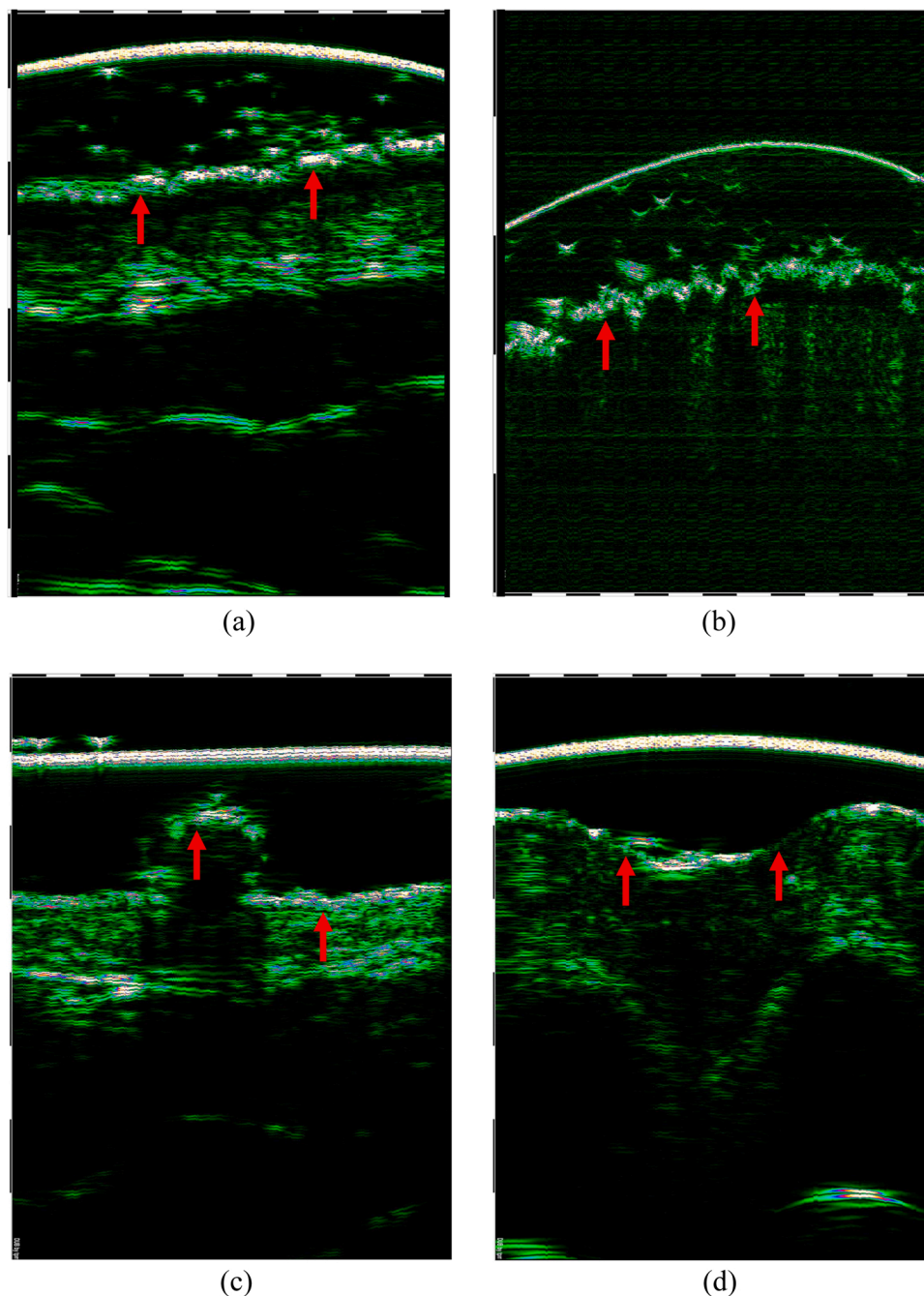


Fig. 1. HFUS (75 MHz) of different skin diseases: atopic dermatitis (a), psoriasis (b), basal-cell carcinoma (c). Red arrows indicate the epidermal layer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

presented workflow successfully segments the epidermal layer and outperforms the state-of-the-art methods in this area, including our previous research (Czajkowska et al., 2021). We also provide the experimental results on the model design, e.g., on selecting the DeepLab v3+ backbone or parameter settings.

The HFUS datasets used for training and validation are specified in Section 2.1. The segmentation framework is presented in detail in Section 2.2. Section 2 describes the training and validation experiments along with the results. We assess our model in various architecture modes and parameter settings. We also compare it to the state-of-the-art approaches. The study is discussed in Section 4, while Section 5 concludes the paper.

2. Materials and methods

2.1. Materials

To design a segmentation algorithm to analyze images of different skin diseases, we combined two HFUS datasets. The first one, further called benchmark, consists of 380 HFUS images of 380 patients with atopic dermatitis (303 images) or psoriasis (77). The dataset is publicly available¹ along with the pre-trained SegUnet model for skin layer segmentation (Czajkowska et al., 2021). The second dataset consists of 200 images of 32 patients with different non-melanocytic skin tumors: BCC (143 images), fibroma (32), skin metastasis of breast cancer (10), keratofibroma (9), superficial BCC (3), and squamous-cell carcinoma (3). Thus, the methodology was designed, trained, and tested using a total of 580 HFUS images.

All the image data were acquired using DUB SkinScanner75 with a 75 MHz transducer. The images have the same size of 2067×1555 pixels, but different resolutions (lateral $\times$ axial): 0.0019×0.085 , 0.0024×0.085 , 0.0031×0.085 , 0.0019×0.085 mm/pix. The images were acquired at the affected skin area. Fig. 1 presents the appearance of the epidermis in different diseases. In each case, the epidermal layer was delineated by one expert and verified by another. Both experts work with HFUS examinations in their daily practice for couple years.

2.2. Methodology

The general scheme of the epidermis segmentation methodology is presented in Fig. 2. The trigger preceding the deep segmentation stage indicates two options we tested during the network design: with either the entire original image or the epidermis region of interest (ROI) being an input image to the main segmentation procedure. The experimental results given in Section 2 favor the application of ROI extraction. Hence, our final segmentation framework involves the upper branch in Fig. 2. Moreover, the final ROI extraction algorithm takes advantage of the DeepLab v3+ segmentation of the full input image. Therefore, for clarity, we describe the DeepLab v3+ segmentation stage first, before the ROI detection.

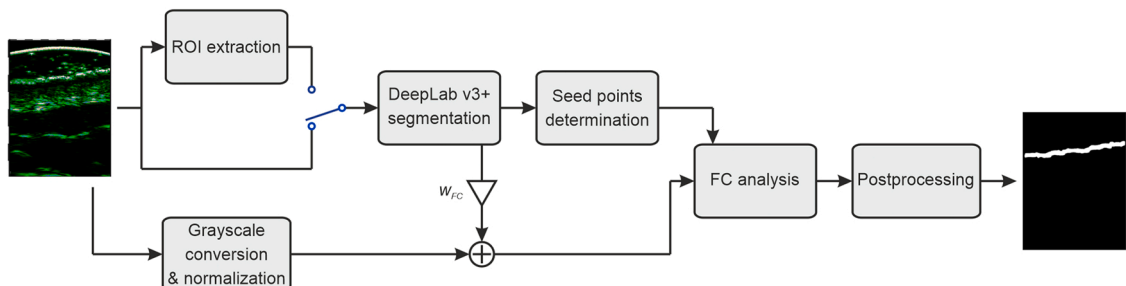


Fig. 2. General epidermis segmentation scheme.

2.2.1. Semantic segmentation using DeepLab v3+ and transfer learning

Regardless of the input image form and size, the main part of the segmentation framework is performed with the DeepLab v3+ network built on a pre-trained backbone for feature extraction (Chen et al., 2017, 2018). Three models were tested to serve as a backbone: ResNet-18, ResNet-50 (He et al., 2016), and Xception (Chollet, 2017). We trained the original models via transfer learning. Originally, the models were pre-trained using the ImageNet database (Deng et al., 2009; ImageNet, 2021). Either the original image or ROI is initially resized to match the size accepted by the DeepLab v3+ (224×224 in both ResNets or 299×299 in Xception). We found the DeepLab v3+ model with a 50-layer residual backbone (ResNet-50) most efficient in a series of experiments. Quantitative inter-model comparison can be found in Section 2.

Our training experiments consistently favored the stochastic gradient descent with momentum optimizer (SGDM) over the adaptive moment estimation (ADAM) optimizer. Thus, SGDM was selected for transfer learning. Learning rates for weights and biases of the convolution and transposed convolution layers in the decoder were enhanced ten times compared to the encoder layers. We used data augmentation through random geometric transformations (horizontal reflection, ± 10 -pixel translation in both directions). The batch size for training was set to 8 and the maximum number of epochs to 200. The loss function employed class weighting to handle possible epidermis/background pixel distribution imbalance. During the experiments, we investigated two loss functions: weighted cross-entropy loss:

$$CEL = -\frac{1}{J} \sum_{j=1}^J \sum_{m=1}^M w_m q_{jm} \ln(u_{jm}), \quad (1)$$

and weighted generalized Dice loss:

$$DL = \frac{1}{K} \sum_{k=1}^K DL_k, \quad (2)$$

where:

$$DL_k = 1 - \frac{2 \sum_{m=1}^M w_m \sum_{j=1}^{J_k} q_{jm} u_{jm}}{\sum_{m=1}^M w_m \sum_{j=1}^{J_k} u_{jm}^2 + q_{jm}^2}. \quad (3)$$

J , K , M denote the total number of pixels, images, and classes ($M = 2$) in the training set, respectively; in DL_k , J_k is the number of pixels in the individual image k . q_{jm} indicates that, according to the ground truth, the j -th pixel belongs to the m -th class, while u_{jm} is the network's decision on the j -th pixel assignment to the m -th class (the output of the m -th map in softmax layer). Weight w_m for the m -th class is inversely proportional to the number of m -th class pixels in the training set.

2.2.2. ROI extraction

The main goal for the ROI extraction was to reduce false-positive regions in the US probe area and other structures of epidermis-like

characteristics. The first approach employed procedures described in Czajkowska et al. (2020, 2021). Yet, the study indicated the requirement of manual user intervention in some cases (e.g., when the US probe is distinctly visible in the image). A series of experiments on the DeepLab v3+ model over the entire input image inspired us to base the ROI detection on this approach. Fig. 3 shows the fully-automated ROI detection and extraction workflow. The coarse DeepLab v3+ segmentation fed with the full input image resized to 224×224 pixels stands for the first part. The specification from Section 2.2.1 applies to this model (with the ResNet-50 backbone and Dice loss). That means that with the ROI extraction, our method uses a cascade of two deep models for segmentation.

Further analysis relies on the network's softmax layer output: a map for the *epidermis* class. The map is (1) binarized through thresholding (zero threshold), (2) morphologically eroded to disconnect regions and remove noisy objects, and (3) subjected to connected component analysis. Each connected component is assigned a label equal to the mean softmax probability of the pixels that constitute it.

Then, to reduce false positives, we analyze the label map column-wise. The generally horizontal arrangement of skin layers sets the assumption that the epidermis object should be the only one present in each column. Thus, only pixels of the object with the largest label are left in a column. The resulting binary image is dilated to restore areas reduced by erosion. Finally, we determine the ROI through the epidermis bounding box with experimentally selected margins: the bounding box is extended 0.5 mm up and 1 mm down. The obtained coordinates enable cropping the ROI from the original image and resizing it to the size required by the DeepLab v3+ model (224^2 or 299^2 , depending on the backbone).

2.2.3. Fuzzy connectedness analysis

Fuzzy connectedness belongs to the region growing image segmentation techniques (Udupa and Samarasekera, 1996; Udupa and Saha, 2003). It is often a part of a broader methodology, mainly used for fine segmentation (Carvalho et al., 1999; Czajkowska et al., 2010). In general, the idea of fuzzy connectedness introduces soft computations to digital spaces. It links pairs of spels (spatial elements; pixels or voxels) with fuzzy relations like affinity κ for topologically adjacent spels and connectedness K for any pair of spels (Udupa and Saha, 2003). The more similar two points in terms of selected characteristics (intensity, intensity gradient), the greater their affinity and connectedness. Application of the FC segmentation requires determination of the fuzzy connectivity scene C_o :

$$C_o(c) = \mu_K(o, c), \quad (4)$$

where: c denotes the spel, o is the seed point, and $\mu_K \in [0,1]$ is the fuzzy connectedness relation between o and c . $C_o(o, o) = 1$ for a seed point. If there are more seed points, then this principle applies to each one. In such a case, the fuzzy connectivity scene is a fuzzy union of scenes determined separately for each seed point.

In this study, we use the canonic affinity function μ_K for two spels c ,

d proposed by Saha et al. (2000) as a combination of Gaussian curves:

$$\mu_K(c, d) = w_1 e^{-\frac{(I(c)+I(d)-m_1)^2}{2\sigma_1^2}} + w_2 e^{-\frac{(|I(c)-I(d)|-m_2)^2}{2\sigma_2^2}}. \quad (5)$$

$I(\cdot)$ denotes the spel intensity in the image. $m_1, \sigma_1, m_2, \sigma_2$ are affinity parameters related to the object characteristics determined based on seed points and their neighborhood. Weights w_1, w_2 balance the intensity and intensity gradient components of μ_K ($w_1, w_2 > 0$ and $w_1 + w_2 = 1$). There are multiple ways to obtain the binary or multiclass segmentation result from the fuzzy connectivity scene C_o (Udupa et al., 2002; Herman and Carvalho, 2001) yet the most common is to apply thresholding (Udupa and Saha, 2003).

In this study, we applied the FC to perform fine segmentation with a set of seed points obtained from the deep model output via a dedicated procedure. We set both weights from 5 to 0.5. To calculate C_o , we used the dynamic programming implementation optimized for matrix-wise processing (Badura et al., 2011). Different thresholds t_{FC} were investigated for C_o binarization with the report given in Section 2. The optimal t_{FC} value was found at 0.1.

Input image for the FC analysis.

The FC analysis involves data from both the original image and the outcome of the deep model (Fig. 4). The input image I_{FC} for the FC is the combination of the ROI of the original image converted to grayscale and normalized to a 0–1 range I_{GN} and the *epidermis* class map from the softmax layer of the network I_{SE} :

$$I_{FC} = I_{GN} + w_{FC} \cdot I_{SE}. \quad (6)$$

The w_{FC} coefficient allows to weigh the effect of the network output map against the original image intensities since both have values from

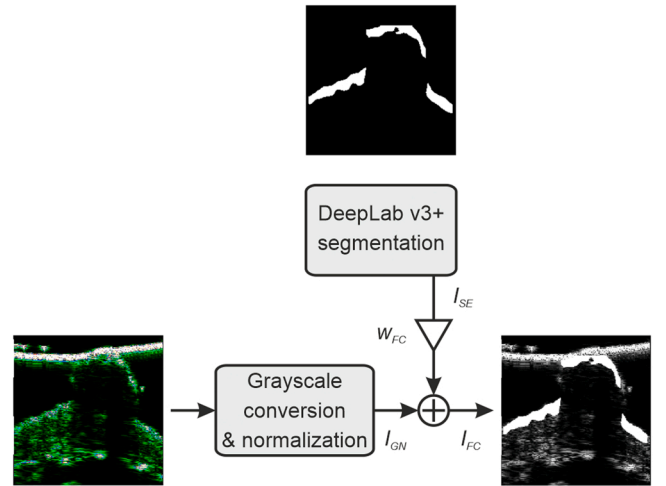


Fig. 4. Preparation of the input image for the FC analysis.

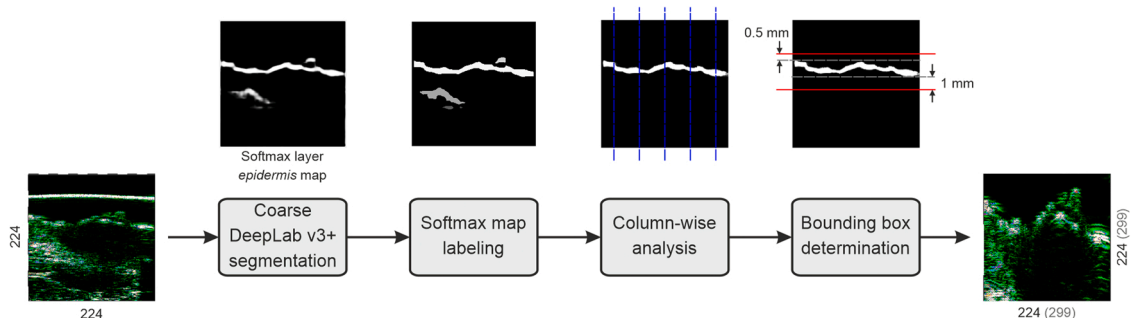


Fig. 3. ROI extraction scheme.

the 0–1 range. Again, a report from experiments on w_{FC} selection with the optimal value found in 1.5 is available in Section 2.

2.2.3. Seed points determination

The seed points for the FC analysis are carefully prepared based on the deep network output to enable fine epidermis segmentation. The procedure (Fig. 5) is, in most parts, similar to the ROI extraction. The *epidermis* class map from the softmax layer of the DeepLab v3+ network is binarized (here 0.5 threshold), eroded, labeled, and analyzed for most significant objects in columns. However, the resulting binary image is not dilated back. Instead, we calculate the Euclidean distance transform within each connected component and remove 25% of pixels with the smallest distances to the background. The remaining core pixels constitute the set of seed points.

2.2.4. Postprocessing

The outcome of the FC scene thresholding has a deep model-dependent size of 224×224 or 299×299 . To smoothly restore the original size, we use nearest-neighbor interpolation supported by blur-ring filtering with a 5×5 kernel and a threshold of 0.5.

2. Experiments and results

The training and experiments were performed in the Deep Learning Toolbox (version 14.2) of the Matlab software (9.10.0.1602886, R2021a) on a workstation with an 8-Core CPU @ 3.20 GHz, 64 GB RAM, and Nvidia Quadro RTX 6000 24 GB GPU.

We used the external 10-fold cross-validation to assess the segmentation. The non-testing remaining data were divided into training and validation subsets (8:1 ratio) in each experiment. Moreover, we ensured that the image data of a single patient from the neoplastic dataset (200 images of 32 patients) are used only for either training or testing, never being spread to both subsets. We also secured disease-wise partition of the dataset. The images were split randomly between folds, but with secured proportions of cases with specific diseases, e.g., 20 out of 200 tumor images were put into each fold, etc. To assess the segmentation, we employed the Dice index, commonly used to measure the spatial overlap in medical imaging studies:

$$D = \frac{2TP}{2TP + FP + FN}, \quad (7)$$

where: TP, FP, FN are the number of true-positive, false-positive, and false-negative epidermis pixels, respectively. In each 10-fold cross-validation, the testing results from folds are gathered and presented using box plots over the entire dataset.

As introduced in Section 2, we verified three different pre-trained models as a backbone for the DeepLab v3+ architecture: ResNet-18, ResNet-50, and Xception. Each of them was trained using two different loss functions: cross-entropy and Dice loss. Moreover, we also analyzed the rationale for the ROI detection procedure. Thus, each

experiment was performed in both deep model input modes: the entire original image and the ROI (note the trigger in Fig. 2). Fig. 6 presents the results of all combinations of those three setup options.

Two observations are evident. First, the ROI detection step improved the segmentation accuracy in all cases. Second, training was more efficient with the Dice loss function. Therefore, in the following experiments, we (1) extracted the epidermis ROI to feed the main DeepLab v3+ model and (2) trained the model using the Dice loss.

In the next experiment, we compared the performance of the full segmentation framework from Fig. 2 with the outcome of the main DeepLab v3+ model. This study aimed to justify the use of the fuzzy connectedness analysis with the seed points determination procedure. The results are shown in Fig. 7.

In each model, the FC-based fine segmentation improved the median Dice index by up to 0.012. The results from Figs. 6 and 7 favored the ResNet-50 over two others. Thus, we selected it to be the ultimate backbone for our DeepLab v3+ architecture.

The experiments so far were performed with parameters $w_{FC} = 1.5$ and $t_{FC} = 0.1$ both found optimal for the segmentation. To justify the parameter settings, we performed additional analysis, presented in Fig. 8.

The investigation involved w_{FC} ranging from 0.25 to 2 with a 0.25 step, and t_{FC} from 0 to 0.95 with a 0.05 step. The interpolated Dice index heatmap is shown in Fig. 8a. The maximum is located at $w_{FC} = 1.5$, $t_{FC} = 0.1$. It can be noticed that the lower contribution of the deep model (w_{FC}), the higher the corresponding threshold providing the optimal segmentation. Fig. 8b and c present the cross-sections of the double-parameter space along the optimal parameter values. The differences in Fig. 8b are relatively slight, yet the analysis favors $w_{FC} = 1.5$. That refers to a relatively significant contribution of the DeepLab v3+ softmax layer map in the input image for the FC analysis. In Fig. 8c, $t_{FC} = 0.1$ brings the clear maximum.

The set of experiments described above let us establish the final shape of our model. Then, we validated it in different skin pathologies: AD, psoriasis, and skin tumor (Fig. 9).

The worst results were obtained in skin tumor, though the median D of 0.85 is still a fair efficiency. In AD and psoriasis, the median Dice index exceeded 0.93.

Finally, to compare our model to the state-of-the-art approaches, we trained our model over the benchmark database¹ (without the skin tumor dataset). We employed the CE-Net (Gu et al., 2019) and two best approaches from (Czajkowska et al., 2021): SegUNet and U-Net. We also applied transfer learning to two other models: FCN-AlexNet and FCN-8 s, pre-trained on CamVid and ImageNet datasets using SGDM and Adam optimizers, respectively. The results are shown in Fig. 10.

They prove the advantage of our multistage methodology based on the DeepLab v3+ and fuzzy connectedness over other models.

We also extended the comparison by the epidermal layer thickness measurements. Fig. 11 shows the boxplots of errors in thickness estimation using a given method and the ground truth. The median error

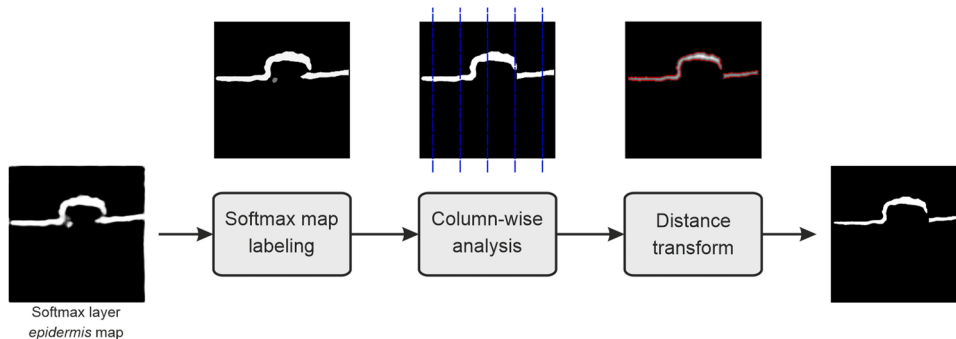


Fig. 5. Seed points determination scheme.

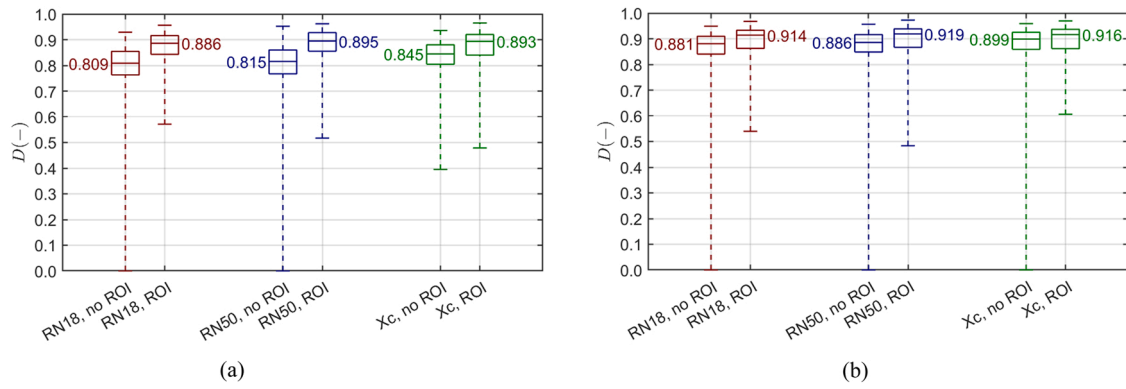


Fig. 6. Dice index box plots using three different models as a DeepLab v3+ backbone in the original image and the ROI mode. Cross-entropy loss (a) or Dice loss (b) was used for training. RN18 – ResNet-18, RN50 – ResNet-50, Xc – Xception. Each box covers 25th to 75th percentile confidence interval with median given and indicated by a central line. Whiskers border the extreme values.

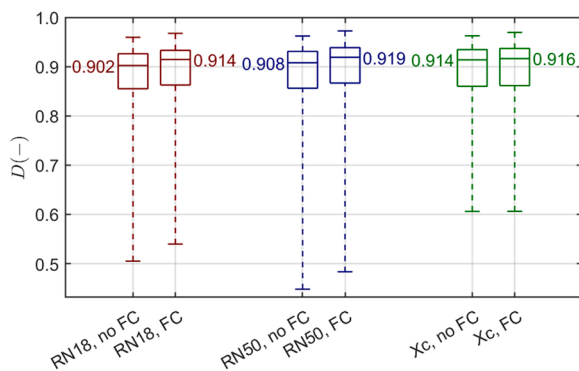


Fig. 7. Dice index box plots using three different models as a DeepLab v3+ backbone with and without applying the fuzzy connectedness analysis. RN18 – ResNet-18, RN50 – ResNet-50, Xc – Xception. Each box covers 25th to 75th percentile confidence interval with median given and indicated by a central line. Whiskers border the extreme values.

produced by our model is at 5 μm , not exceeding $\pm 75\mu\text{m}$ in any case. Finally, we estimated the time consumed by our final model per case during the analysis to be 0.47 ± 0.02 s, with 0.16 s taken by both DeepLab v3+ models in total and 0.07 s by the fuzzy connectedness analysis.

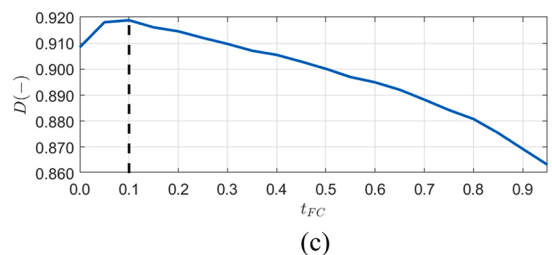
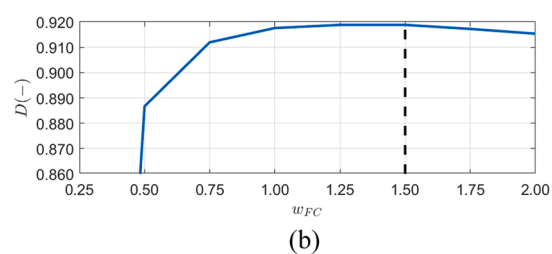
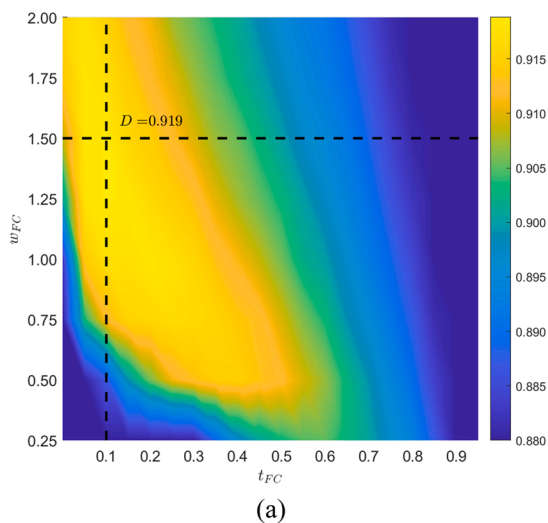


Fig. 8. Median Dice index vs. t_{FC} and w_{FC} . Heatmap vs. both parameters is shown in (a) with the maximum D given for $t_{FC} = 0.1$ and $w_{FC} = 1.5$. (b) and (b) present the cross-sections along optimal arguments: the D distribution vs. the w_{FC} coefficient at $t_{FC} = 0.1$, and vs. the t_{FC} threshold at $w_{FC} = 1.5$, respectively.

4. Discussion

The proposition of a novel framework for epidermis segmentation in different skin diseases based on 75 MHz HFUS brings several points to the discussion. We performed experiments on the model components, including loss function, region of interest extraction, deep network backbone, or fuzzy connectedness adjustment. For segmentation quality

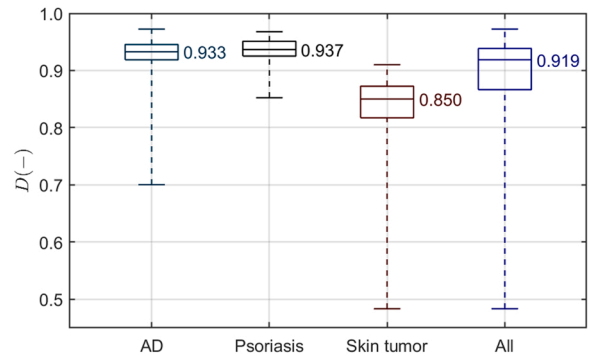


Fig. 9. Dice index box plots in different types of skin pathologies. Each box covers 25th to 75th percentile confidence interval with median given and indicated by a central line. Whiskers border the extreme values.

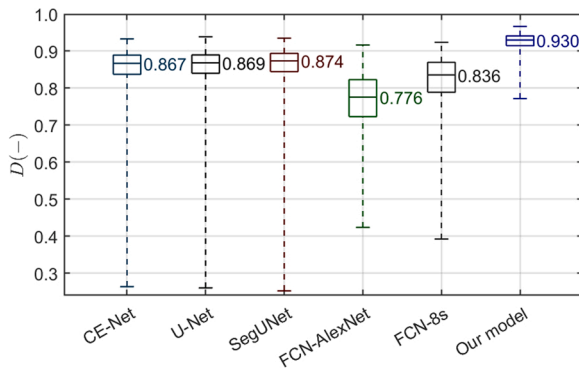


Fig. 10. Dice index box plots using our model and state-of-the-art methods over the benchmark dataset (AD and psoriasis cases). Each box covers 25th to 75th percentile confidence interval with median given and indicated by a central line. Whiskers border the extreme values.

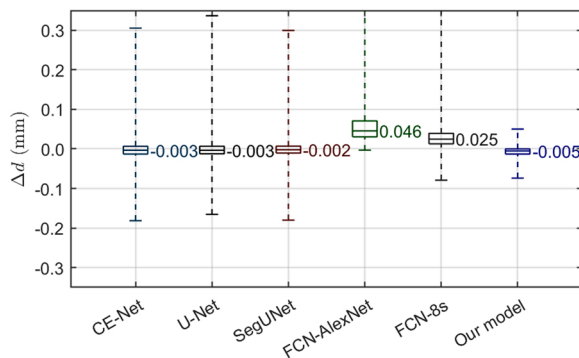


Fig. 11. Epidermis thickness errors box plots using our model and state-of-the-art methods over the benchmark dataset (AD and psoriasis cases). Each box covers 25th to 75th percentile confidence interval with median given and indicated by a central line. Whiskers border the extreme values.

assessment, we involved the Dice coefficient due to its well-established position in the medical image domain. It also enabled reliable comparison to other epidermis segmentation models or methods described in the literature using the same benchmark dataset¹. Deep learning software designers often employ the IoU measure (Jaccard index) for validation. For the reasons given above and since both measures are recalculable using a simple formula, we decided not to provide IoU results.

To put our study into perspective, we took or trained five different deep learning models recognized in analyzing HFUS or OCT images. Three of them were trained from scratch, while two others through transfer learning, all over the benchmark database containing inflammatory disease cases.

The CE-Net network, being initially developed by Gu et al. (2019) for the OCT image segmentation, was selected assuming the similarity to OCT images by a characteristic layered structure. We supported CE-Net by the FCM clustering and obtained a Dice index of 0.867. The second approach with the core U-Net architecture given by Ronneberger et al. (2015), also improved by the FCM-based preprocessing, produced $D = 0.869$. The third model, SegUNet designed by Czajkowska et al. (2021) yielded the highest efficiency in this group with $D = 0.874$.

The most critical concern of this analysis is the training part. All the methods mentioned above were trained from scratch, whereas the current approach is built on the DeepLab v3+ backbone pre-trained on the ImageNet database. As we expected, transfer learning with a relatively small dataset significantly improved the segmentation results up to $D = 0.919$ over all three skin diseases and $D = 0.930$ over the inflammatory dataset. Hence, we also adjusted two other pre-trained models,

widely described in the medical image segmentation literature (Morid et al., 2021): FCN-AlexNet and FCN-8 s, pre-trained on CamVid and ImageNet datasets, respectively. The validation results significantly lower than those produced by our model in different setups and parameter settings proved the DeepLab v3+ model superiority.

A look at the epidermal layer thickness errors suggests that the deep learning models under consideration offer fair accuracy in this matter. However, our DeepLab v3+ model stands out among them with the smallest dispersion of errors, not exceeding ± 75 micrometers.

Another concern was the influence of the ROI selection for the segmentation outcome. Comparative analysis in this matter was skipped in the previous work (Czajkowska et al., 2021), where the quantitative experiments were limited to the framework relying on already detected ROIs. In our current work, we designed a new DeepLab v3+ driven ROI detection step to deal with the sharp appearance of the US probe head. The design is supported by experiments over models employing the ROI extraction or not. The ROI detection step significantly improves the segmentation. It is mainly connected with the information loss when resizing high-resolution input images to the 224×224 size. The loss is more significant in original data than cropped ROI. The problem is also addressed by the smoothing filter accompanying the final mask interpolation in postprocessing. Moreover, input image reduction limits the impact of image areas of similar appearance to the epidermis layer, causing the increase of false-positive segmentations (note that the FC-based adjustment also addresses this problem). Some examples of the ROI detection influence on the results are visualized in Fig. 12 (middle images), where the results obtained with and without this step are given in red and blue, respectively.

The vital contribution of our work is the fuzzy connectedness-based fine segmentation with a dedicated procedure for the seed region selection. The obtained results show that this step noticeably improves the segmentation results in all network configurations under consideration. The exemplary results with and without the FC adjustment are shown in Fig. 12 (rightmost images), where the segmentation results obtained using DeepLab v3+ with and without the FC-based fine segmentation are outlined in red and green, respectively.

The observations made during the study suggest that the FC analysis with accompanying procedures improves the deep model output twofold. On the one hand, careful selection of seed points through pruning the network outcome limits false positive areas. On the other hand, the FC scene determination and thresholding reduce the number of false negatives. Including the network softmax map in the input image has a noticeable effect on the latter. US images are so difficult to analyze (noise, distortions, etc.) that relying solely on the raw data does not bode well to follow fuzzy relations between spels. Our research confirms the above statement. Adding the fuzzy pixel probability map from the previous step to the FC input image improves the ultimate segmentation reliability and efficiency (Fig. 8a).

The limitation of the study can be found in the data acquired by a device of one type, with a single frequency of 75 MHz, optimal in imaging diseases under consideration. However, only one vendor on the market offers this frequency. Two vendors provide 50 MHz probes, which are also used in clinical practice. The applicability of our models to the images acquired at 50 MHz can become a point of interest since the skin appearance is relatively similar in both cases. Even though a direct application can be complex, perhaps the 75 MHz models can easily be retrained, e.g., through transfer learning, to work with the 50 MHz data. We also find possible employment of coarse segmentation results produced by our first deep model as auto-context information for the second model, so the framework could be trained end-to-end, like in some recent works on the HFUS analysis by Xu et al. (2020) and Qiu et al. (2021).

Since HFUS appeared in daily dermatological practice in the last few years, access to image data described and delineated by an expert is rare. However, compared to our previous research (Czajkowska et al., 2020, 2021), the current analysis is based on the extended database. Moreover,

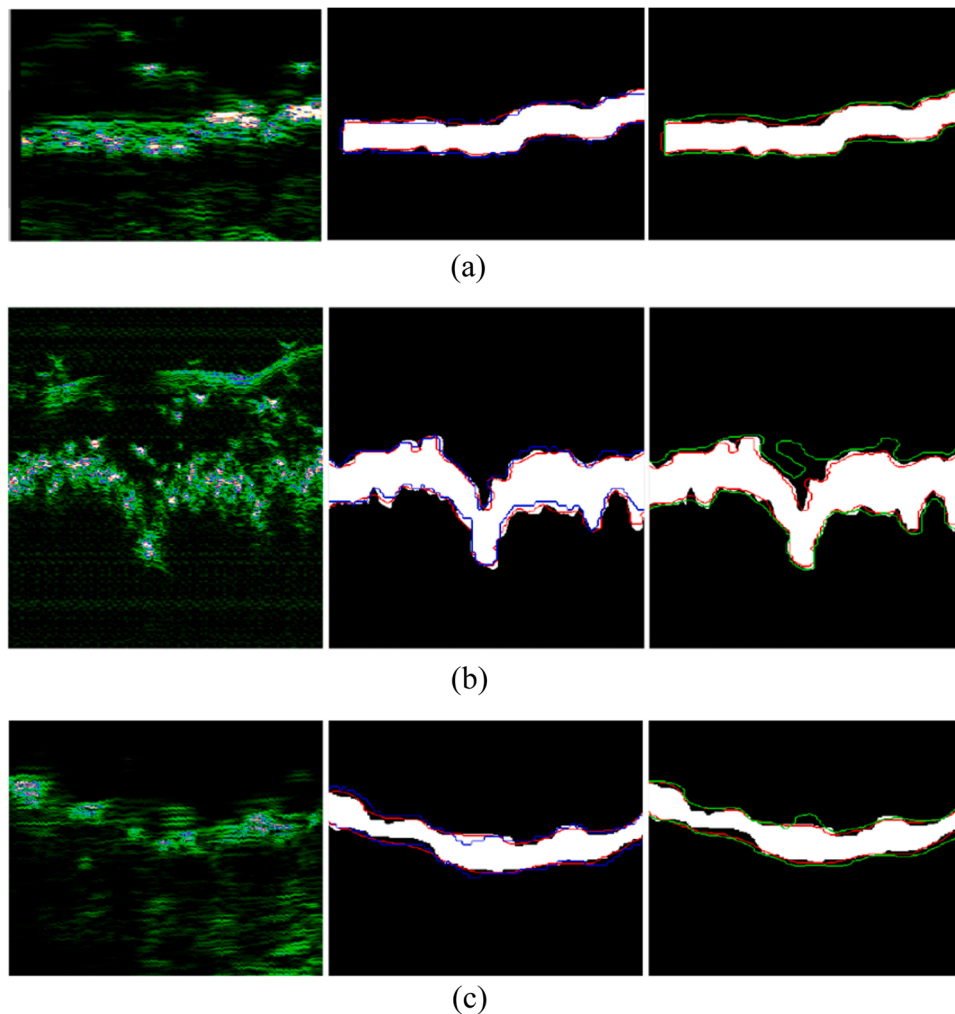


Fig. 12. Exemplary segmentation results in AD (a), psoriasis (b), and skin tumor (c). Left to right: original image ROI, illustration of ROI extraction impact, and FC analysis impact. The white area shows the ground truth mask. Red contour refers to the ultimate model outcome. Blue and green contours delineate the segmentation results without the ROI extraction and without applying the FC-based fine segmentation, respectively.

we keep working to include more pathologies in the dataset. We also do our best to fight the database size, which might be questioned for deep learning purposes, e.g., through data augmentation or transfer learning. We assume that future research with the extending dataset should yield comparable or even better segmentation accuracy.

5. Conclusion

The fast development of imaging techniques followed by computer-aided diagnostic tools still leaves room for research in HFUS image analysis of the skin. Our study introduces a fully automated cascade deep learning framework with fuzzy connectedness support for fast, robust, and accurate segmentation of the epidermal layer. The presented framework is our proposition for linking deep learning and fuzzy image analysis methods through the successful application of fuzzy connectedness. According to the experimental results, it is efficient in both inflammatory and neoplastic skin lesions. The obtained segmentation and epidermis thickness estimation accuracies justify the application of our approach in computer-aided skin diagnosis.

CRedit authorship contribution statement

Joanna Czajkowska: Conceptualization, Methodology, Software, Validation, Investigation, Resources, Writing – original draft, Writing –

review & editing, Visualization, Project administration. **Paweł Badura:** Validation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition. **Szymon Korzekwa:** Resources, Data curation. **Anna Płatkowska-Szczerek:** Conceptualization, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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